Project 6 ML4T
Indicators and TOS

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Introduction:

This report will evaluate five technical indicators which will be used in the following projects to assess the effectiveness in stock trading. The indicator chosen and analysied in this report are: MACD, BBP, RSI, Momentum and Stochastic Oscillator. Furthermore, this report also includes an analysis of the Theoretically Optimal Strategy (TOS) experiment.

Indicators:

JPM is used as the example stock and benchmark to conduct our analysis. The experimental time range is from 2008-01-01 to 2009-12-31.

MACD:

Concept:

The Moving Average Convergence Divergence (MACD) is used to identify trends and guider trader to buy/sell in trading market. It is a momentum-based technical indicator.

The MACD line is generated based on the different between two Exponential Moving Averages (EMAs):

$$MACD = EMA(12\ days) - EMA(26\ days)$$

And a Signal Line is 9-day EMA of MACD, it represent a smoothed fluctuation of MACD:

$$Signal\ Line = EMA(9days\ MACD)$$

To better analysis the position of MACD and Signal, here we introduce MACD Histogram, which represents the momentum shifts:

$$Histogram = MACD - Signal\ Line$$

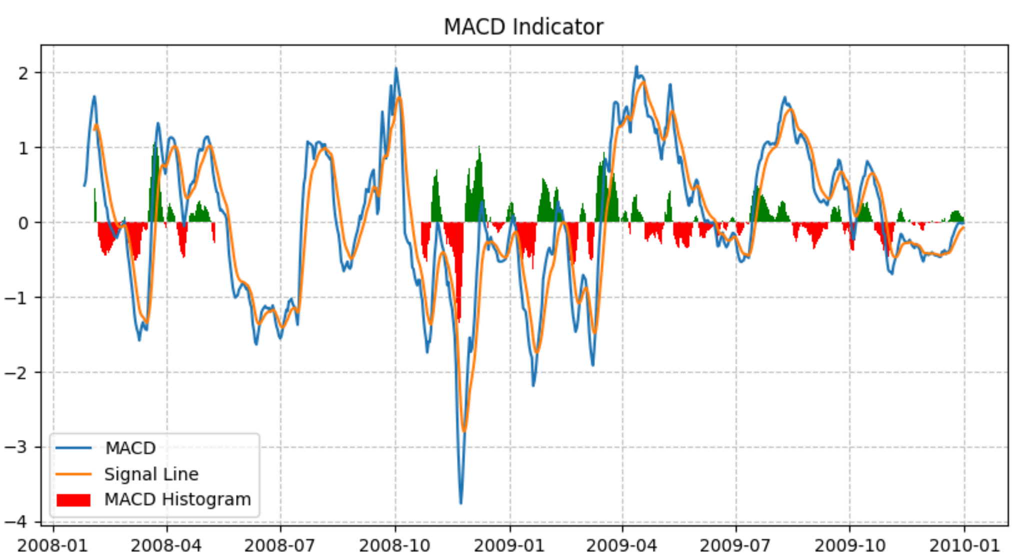
Usage:

When MACD above Signal line ⇒ Histogram is positive ⇒ BUY

When MACD below Signal line ⇒ Histogram is negative ⇒ SELL

Discription:

The chart MACD Indicator displays the MACD line, Signal Line and Histogram. The chart clearly highlight the bullish and bearish momentum. When histogram is green, its a buy opportunities and when it is red, its a sell opportunity. The indicator will works best in trending markets and should also combine with other indicators like RSI or BB to prove the accuracy.



BBP:

Concept:

Bollinger Bands Percent measures the current prices position compared with the Bollinger Bands. Bollinger Bands is the used to represent the volatility and potential overbought and over sold conditions.

$$BBP = (\text{Upper Band} - \text{Lower}) / (\text{BandPrice} - \text{Lower Band})$$

where:

- Upper band = SMA + (2 x Standard Deviation)
- Lower Band = SMA + (2 x Standard Deviation)
- SMA (Simple Moving Average) = Sum(prices).mean()

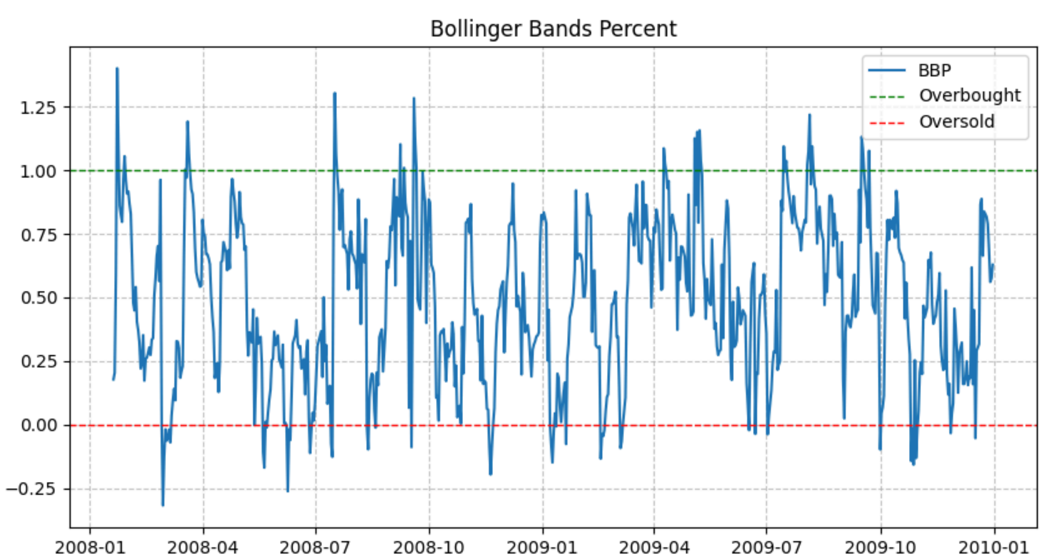
Usage:

When BBP > 1 ⇒ This is a Overbought Region, signal for SELL

when BBP < 0 ⇒ This is a OVersolde Region, signal for BUY

Description:

When a stock has a high BBP value, it suggests that the stock might be to strong in the higher value and a possible price drop might happen, and vice versa. In below chart Bollinger Bands Percent, the Green line is the overbought threshold and red line represent for the over sold threshold.



RSI:

Concept:

Relative Strength Index (RSI) represents the speed and magnitude of recent prices changes to determine if a stock is overbought or oversold. It ranges from 0 to 100.

$$RSI = 100 - 100 / (1 + RS)$$

where:

RS (Relative Strength) = the average return in N days / average loss in N days

In this experiment, we use 14 days as the calculation period.

Usage:

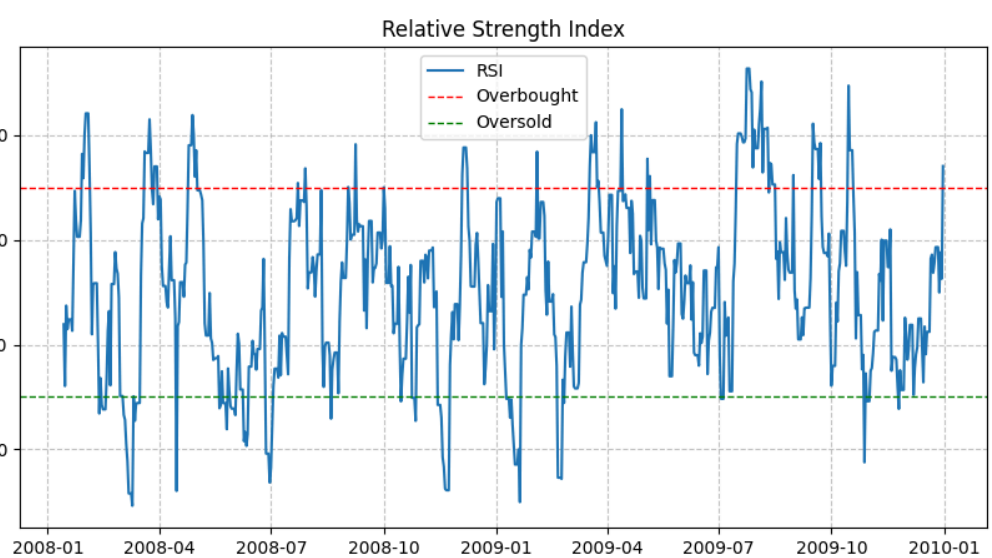
If RSI > 70 ⇒ Overbought position, it might be a sell signal.

if RSI < 30 ⇒ Oversold position, it might be a buy signal

RSI between 30 ~ 70 ⇒ Netural

Description:

RSI is a powerful momentum indicator to help traders to identify overbought and oversold. When combining with MACD and BBP, this indicator will provide better confirmation for a buy or sell signals.



Momentum:

Concept:

Momentum Indicator (MOM) analysis the rate of changes and in current stock over the past specific period. This is a kind of trend-following oscillator. MOM will help traders to identify potential reversals and trend strength.

$$MON = Price(t) - Prices(t - N)$$

where:

Price(t) = current closing price

Price(t - N) = closing price N days ago

In this experiment, we take N = 14

Usage:

if MOM > 0 ⇒ means the bullish momentum. It is a BUY signal

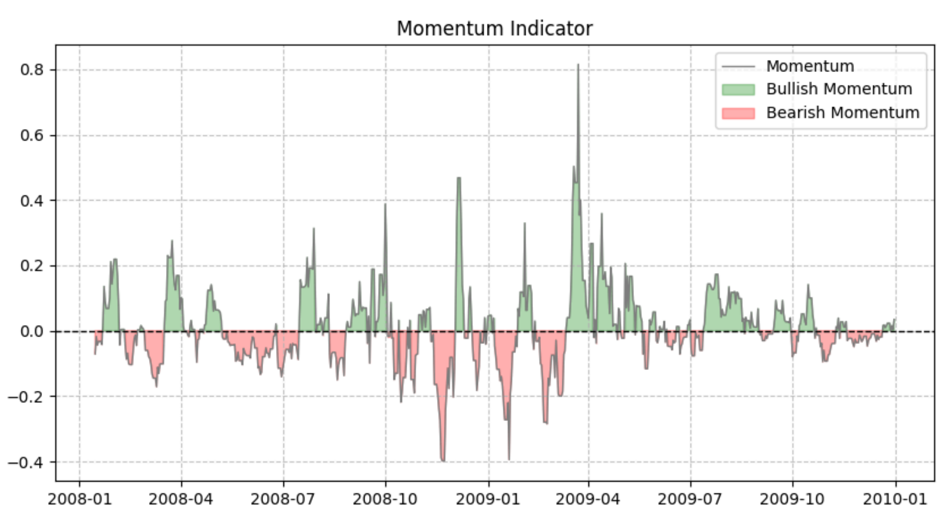
if MOM < 0 ⇒ means the bearish momentum, it is a SELL signal

If prices is rising but MOM is falling ⇒ might be a weakening uptrend

If price is falling but MOM is rising ⇒ might be a trend reversal

Description:

A higher MOM value indicates a strong trends. and wehn MOM starts declining after the peak, it may indicates that there might be a loss of momentum and reversal. Combining with MACD or RSI, MOM will better validate bullish or bearish trends. According to the charts, Green shaded areas are bullish momentum and red si bearish momentum.



Stochastic Oscillator :

Concept:

Stochastic Oscillator (SO) indicator compares the stocks closing price with its price range over a certain period, often 14 days. This indicator helpers to determine if a stock is overbought or oversold.

%K = (Price - Lowest Low) / (Highest High - Lowest Low) * 100

where:

- Price = Most recent closing price.
- Lowest Low = Lowest price over the past N periods.
- Highest High = Highest price over the past N periods.
- Common value for N is 14 periods
- %K = the current value of stochastic indicator

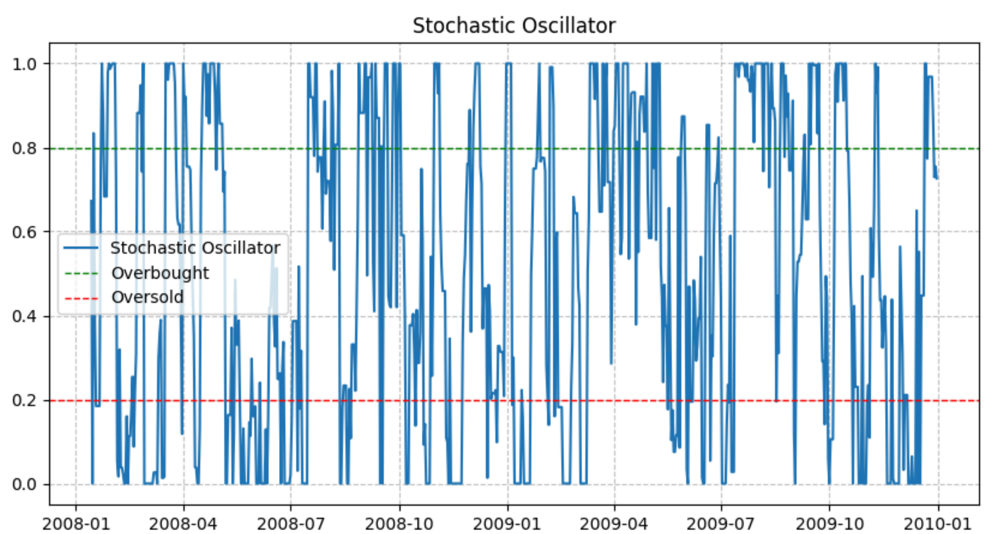
Usage:

if %K > 0.8 ⇒ overbought zone, a potential SELL signal

if %K < 0.2 ⇒ oversold zone, a potential BUY signal

Description:

Stochastic Oscillator is a indicator to identify overbought and oversold conditions. It's range is from 0 to 1. As the chart shows, the red line and green line representing the overbought and oversold threshold. When the %K combined with trending indicator such as MACD or RSI, it can better define the trade decisions.



Theoretically Optimal Strategy (TOS):

Strategy:

The Theoretically Optimal Strategy (TOS) is a experiment trying to adopt a optimal strategy to maximize the stock investment return. The strategy is buying at local minimum and selling at local maximum based on the known future price data. In this experiment, we assume we have a perfect foresight so we could maximize the potential profit.

Below is the strategy utilizing perfect foresight:

```
if next_price > current_price:
    BUY all shares
Else:
    SELL all shares
```

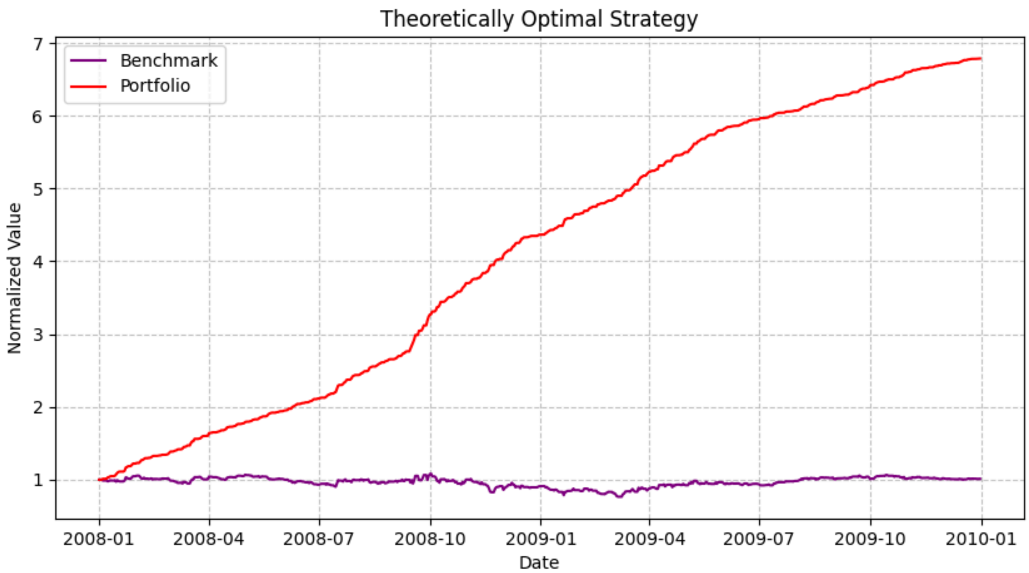
Performance Comparison:

The TOS strategy significantly beat the performance of benchmark (JPM). The below table shows the comparison of matrix between benchmark and TOS portfolio.

Metric	Benchmark	TOS Portfolio
Cumulative Return	0.012300	5.786100
Stdev of Daily Returns	0.014125	0.004170
Mean of Daily Returns	0.000116	0.002635

The TOS achieves a 5.7X return while Benchmark only has 1.2%. TOS also have a lower standard deviation which indicates its more table than benchmark. The Mean daily return of TOS is also much higher than Benchmark.

The chart Theoretically Optimal Strategy visually display the experiment result and showing that how a best case scenario can be when people know the future prices.



Theoretically Optimal Strategy
Benchmark
Portfolio
Normalized Value
Date
2008-01
2008-04
2008-07
2008-10
2009-01
2009-04
2009-07
2009-10
2010-01